Shubhadeep Mandal

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Education

International Institute of Information Technology, Hyderabad — July 2025 – Present
Telangana, India
Master of Technology in Computer Science and Engineering — *CGPA: 9.12*

Indian Institute of Information Technology, Ranchi — Aug 2020 – Jun 2024
Ranchi, India
Bachelor of Technology (Hons.) in Computer Science and Engineering — *CGPA: 9.05*

Projects

IndicLLM: Transformer Pretraining & Reasoning SFT | *PyTorch, FlashAttention-2, RoPE*

- Pretrained 25M-param decoder-only Transformers from scratch on 1B+ tokens in Hindi and Assamese using bare PyTorch primitives, achieving +36.5% perplexity reduction via FlashAttention-2, RoPE, and SwiGLU.
- Trained 16k SentencePiece BPE tokenizers on AI4Bharat Sangraha, eliminating token fertility tax and achieving 0.505 Bits-Per-Byte compression across high and low-resource languages.
- Fine-tuned 8 model variants across 20k synthetic multi-hop reasoning tasks, demonstrating a +45.8% relative Token F1 gain and 61.3% decomposed CoT step score over direct SFT.

AmbiStory: NLP Plausibility & Disambiguation Engine | *Python, PyTorch, Transformers, NLI*

- Achieved the best non-LLM solution on the AmbiStory problem (SemEval 2026), reaching $\rho = 0.535$ and 72.2% test accuracy.
- Engineered a 202-dim vector fusing multi-context sentence embeddings, NLI priors, and PCA interactions into an ordinal regression head.
- Boosted generalization by +0.265 ρ across unseen homonyms using group-stratified 5-fold CV, annotator KL divergence loss, and weight averaging.

Football Video Analytics with YOLO | *Python, YOLOv8, OpenCV, Optical Flow, K-Means*

- Fine-tuned YOLOv8 for multi-object tracking of players, ball, and referees, with K-Means color clustering for team assignment.
- Compensated camera pan/tilt using Lucas-Kanade optical flow and mapped pixel coordinates to pitch meters via homography.
- Engineered continuous speed, distance covered, and team possession telemetry pipelines with real-time video overlay generation.

Peer-to-Peer Distributed File Sharing System | *C++, Sockets, Multithreading, OpenSSL*

- Built a BitTorrent-style P2P system in C++, achieving an 8.5x speedup on 1GB transfers (193s to 22.8s) across 5 parallel peers.
- Engineered out-of-order parallel chunk writing via `pwrite()` and dynamic bitfield generation for simultaneous partial seeding.

Technical Skills

Languages: Python, C/C++, TypeScript, JavaScript, SQL, Bash.
NLP & Deep Learning: Custom Transformers (Pretraining & SFT), FlashAttention-2, RoPE, SwiGLU, RMSNorm, CoT, DeBERTa, MPNet, PyTorch, Hugging Face, SentencePiece.
Computer Vision & ML: YOLOv8, OpenCV, TensorFlow.js, Scikit-learn, Optical Flow, K-Means, Streamlit.
Systems, Data & Tools: POSIX Sockets, Linux/Unix, Git, Docker, NumPy, Pandas, Matplotlib.

Achievements

- **B.Tech Rank Holder:** Ranked 2nd in IIIT Ranchi by GPA in B.Tech CSE.
- **PGEE 2025:** Secured All-India Rank 100 in IIIT-Hyderabad Post Graduate Entrance Examination (PGEE).
- **Megathon 2025:** Third rank team in Megathon 2025 in the mobile computing track.

Interests

- Web Novel and Manga Enthusiast.
- Chess and Badminton.